

Customer Engagement & Product Utilization Analytics for Retention Strategy

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Abstract

This study analyzes 10,000 customers of a European retail bank to evaluate whether behavioral engagement provides a more actionable churn signal than financial wealth. Using activity status, product ownership, and account balance, customers are classified into engagement profiles and five retention-focused KPIs are constructed. The analysis finds that active customers churn at roughly half the rate of inactive customers regardless of wealth level, and that inactive, high-balance customers represent the bank's largest silent-churn risk segment. A separate, unexplained finding shows that customers holding three or more products exhibit near-total churn rates, challenging the assumption that greater product depth necessarily corresponds to greater loyalty. These findings support a shift in retention strategy from wealth-based targeting toward engagement-based intervention.

1 Introduction

Retail banks have traditionally used financial indicators – account balance, salary, product count – as proxies for customer loyalty, on the assumption that wealthier or more product-engaged customers are less likely to leave. In practice, this assumption often fails: high-balance customers can quietly disengage and eventually churn without triggering any alert, since standard financial monitoring does not track behavioral signals such as account activity.

This project investigates customer churn through the lens of engagement and product utilization rather than demographics or wealth alone. The goal is to determine which behavioral signals most reliably predict churn, identify disengaged-yet-valuable customers before they leave, and translate these findings into concrete retention recommendations.

2 Problem Statement

Despite holding detailed customer data, the bank lacks a clear, quantified answer to three questions:

- Does customer engagement meaningfully predict churn, independent of financial standing?
- Does holding more banking products reduce the likelihood of churn?
- Can disengaged, high-value customers be identified and targeted before they leave?

Without this insight, retention efforts risk being generic – targeting customers by balance tier rather than by actual risk of leaving.

3 Methodology

3.1 Dataset

The dataset comprises 10,000 customer records with 13 usable fields after validation, including credit score, geography, demographic details, tenure, account balance, product count, card ownership, activity status, salary, and churn outcome (**Exited**). The dataset represents a single-year snapshot; a **Year** column present in the raw file was constant across all records and excluded from analysis.

3.2 Data Validation

Prior to analysis, the dataset was checked for missing values, duplicate records, invalid binary encodings, and implausible numeric ranges. No data quality issues were found – all fields were complete, uniquely keyed by **CustomerId**, and within realistic bounds. The overall churn rate of **20.37%** was established as the baseline against which all subsequent subgroup comparisons are made. Notably, 36.2% of customers hold a zero account balance; this was treated as a legitimate customer segment (e.g., a card-only relationship) rather than missing data.

3.3 Engagement Classification

Customers were segmented into four engagement profiles using activity status, product count, and balance relative to the sample median:

- **Active Engaged** – active member, 2+ products
- **Active Low-Product** – active member, 1 product
- **Inactive Disengaged** – inactive member, below-median balance
- **Inactive High-Balance** – inactive member, above-median balance

This classification prioritizes activity status first, then splits each branch by product depth or balance, ensuring every customer receives exactly one profile with no overlap.

For this analysis, a **sticky customer** is defined descriptively as an active customer holding exactly two products, corresponding to the lowest observed churn segment (7.6%, Section 4.3). This is a descriptive label based on observed outcome, not a claim that two products causally produce loyalty. The primary operational **engagement threshold** identified in this analysis is activity status itself: active members consistently exhibit lower observed churn than inactive members across every balance quartile examined (Section 4.4), making activity status the most consistent behavioral line found in the data, though this is an empirical pattern rather than a statistically derived cutoff.

3.4 KPI Construction

Five metrics were defined to quantify retention behavior: Engagement Retention Ratio, Product Depth Index, High-Balance Disengagement Rate, Credit Card Stickiness Score, and Relationship Strength Index.

The **Product Depth Index** is computed by first assigning each product-count tier a retention score of $1 - \text{churn rate}$, then taking the customer-count-weighted average of these scores across all four tiers. Because the underlying churn-by-product relationship is strongly non-monotonic

(Section 4.3), this single index is only meaningful when read alongside the tier-level breakdown, and should not be interpreted on its own as "more products is better."

The **Relationship Strength Index** is a weighted composite using analyst-defined weights: $0.4 \times \text{IsActiveMember} + 0.3 \times \text{ProductFit} + 0.1 \times \text{HasCrCard} + 0.2 \times \text{NormalizedTenure}$. *ProductFit* is a hand-scored value reflecting the retention pattern found in Section 4.3: 2 products scores 1.0 (the empirically most loyal tier), 1 product scores 0.5, and 3–4 products score 0.0 (the empirically least loyal tier). Tenure is normalized to $[0, 1]$ by dividing by the maximum observed tenure (10 years). The weights themselves were chosen by the analyst to reflect the relative strength of each factor observed elsewhere in this report (activity status having the largest single-variable effect among behavioral factors) and are not derived from a statistical model; they should be understood as a documented judgment call rather than a fitted result.

Both indices were sanity-checked prior to use rather than formally validated: the Relationship Strength Index was checked for a negative correlation with churn and for consistency with the geographic churn pattern established in Section 4.1, and both checks passed (Section 4.5).

4 Findings

4.1 Exploratory Patterns

Initial single-variable analysis revealed substantial variation in churn across geography, gender, and age (Figure 1, Figure 2).

- **Geography:** Germany shows the highest churn rate at 32.4%, well above the 20.37% baseline. Spain (16.7%) and France (16.2%) both sit below baseline.
- **Gender:** Female customers churn at 25.1% versus 16.5% for male customers – an 8.6 point gap.
- **Age:** Churn rises sharply through middle age, peaking at 51.1% in the 46–60 band, before dropping to 24.8% for customers 61 and older. The 18–30 band churns least, at 7.5%.
- **Correlation:** Age shows the strongest correlation with churn ($r = +0.29$), followed by *IsActiveMember* ($r = -0.16$) and Balance ($r = +0.12$). Product count, credit score, tenure, card ownership, and salary are all weakly correlated ($|r| < 0.05$).

A preliminary test compared the churn spread produced by activity status against the spread produced by balance level: *IsActiveMember* produced a spread of 0.126 (inactive 26.9% vs. active 14.3%), versus 0.092 for high/low balance (25.0% vs. 15.8%). This first pass already favored engagement over wealth as a churn driver, though neither approached the magnitude of the age effect.

4.2 Engagement Classification Results

The four-profile classification produced a far sharper churn gradient than any single variable in isolation (Figure 3, Table 1).

Active Engaged customers churn at less than a third of the rate of Inactive High-Balance customers. Critically, Inactive High-Balance customers churn *more* than Inactive Disengaged customers – high account balance does not protect a disengaged customer from leaving, and may even correlate with elevated risk.

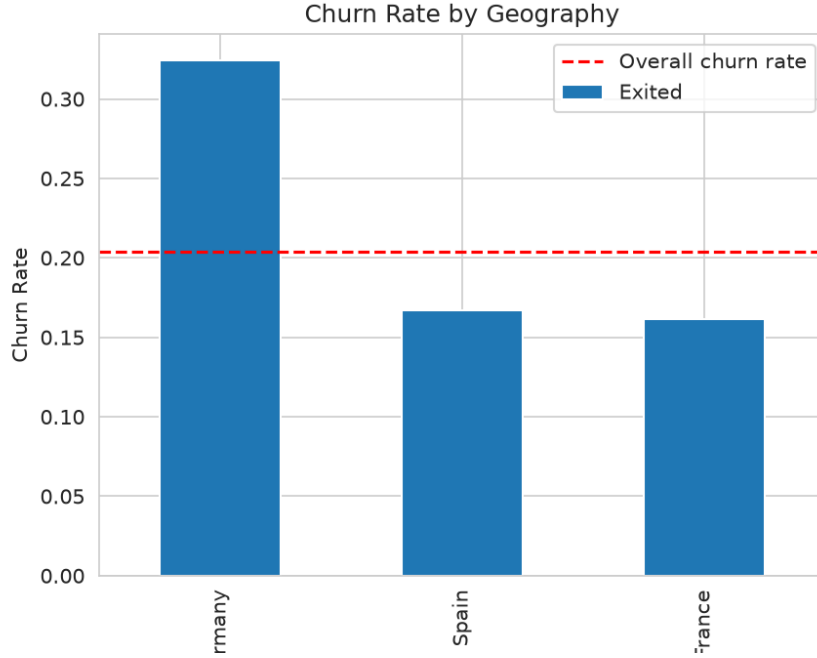


Figure 1: Churn rate by geography, relative to the 20.37% baseline.

Table 1: Churn rate by engagement profile

Profile	Share of Base	Churn Rate
Active Engaged	25.9%	9.7%
Active Low-Product	25.6%	18.9%
Inactive Disengaged	23.9%	21.2%
Inactive High-Balance	24.6%	32.3%

A confounder check found that age was evenly distributed across all four profiles, ruling it out as a distorting factor. Geography was not evenly distributed: the Inactive High-Balance profile is 41.9% German, versus 9.7–24.5% German in the other three profiles. Since Germany’s standalone churn rate (32.4%) closely matches this profile’s rate (32.3%), part of the observed effect likely overlaps with the geography effect rather than being purely behavioral. This is treated as an important caveat rather than a refutation of the finding.

The Inactive High-Balance segment – 2,456 customers, 24.6% of the base, averaging €131K in balance and €101K in salary – represents the disengaged-yet-high-value customer group this project specifically set out to identify.

4.3 Product Utilization Results

Churn rate by exact product count reveals a striking, non-monotonic pattern (Figure 4, Table 2).

Two-product customers are the most loyal segment in the entire dataset (7.6% churn). However, customers holding three or four products churn almost universally, despite comprising only 3.3% of the base. A binary single-vs-multi-product comparison (27.7% vs. 12.8%) obscures this pattern

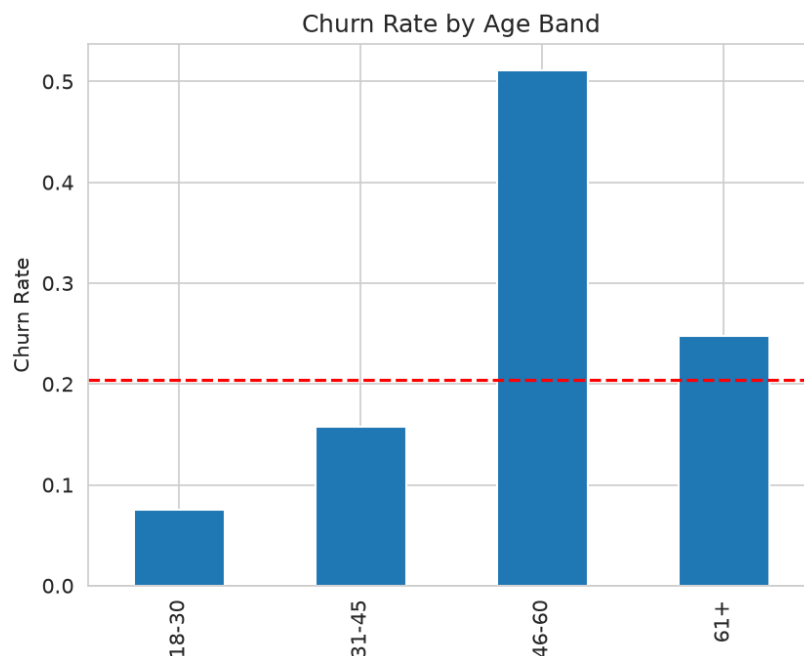


Figure 2: Churn rate by age band, peaking sharply in the 46–60 group.

Table 2: Churn rate by number of products held

Products	Customers	Share of Base	Churn Rate
1	5,084	50.8%	27.7%
2	4,590	45.9%	7.6%
3	266	2.7%	82.7%
4	60	0.6%	100.0%

by merging the most loyal group (2 products) with the least loyal (3–4 products); exact product count, not a binary tier, is the meaningful cut.

The inactivity hypothesis – that 3–4 product customers churn because they are also disproportionately inactive – was tested and does not hold: only 56.4% of this group is inactive, compared to roughly 50% baseline, far too small a gap to explain a churn rate approaching 100%. A similar check against geography (40% German among 4-product customers, versus a 25% base rate) also falls short of explaining the effect. This pattern is therefore flagged as an open question for further investigation – possibly linked to forced product bundling or dissatisfaction with an over-sold relationship – rather than a solved finding.

Credit card ownership showed no meaningful effect on churn (20.8% for non-holders vs. 20.2% for holders), indicating it is not a useful retention lever on its own.

4.4 Financial Commitment vs. Engagement

Cross-referencing balance quartile against activity status confirmed that engagement is protective at every wealth level: active members churned less than inactive members within every balance

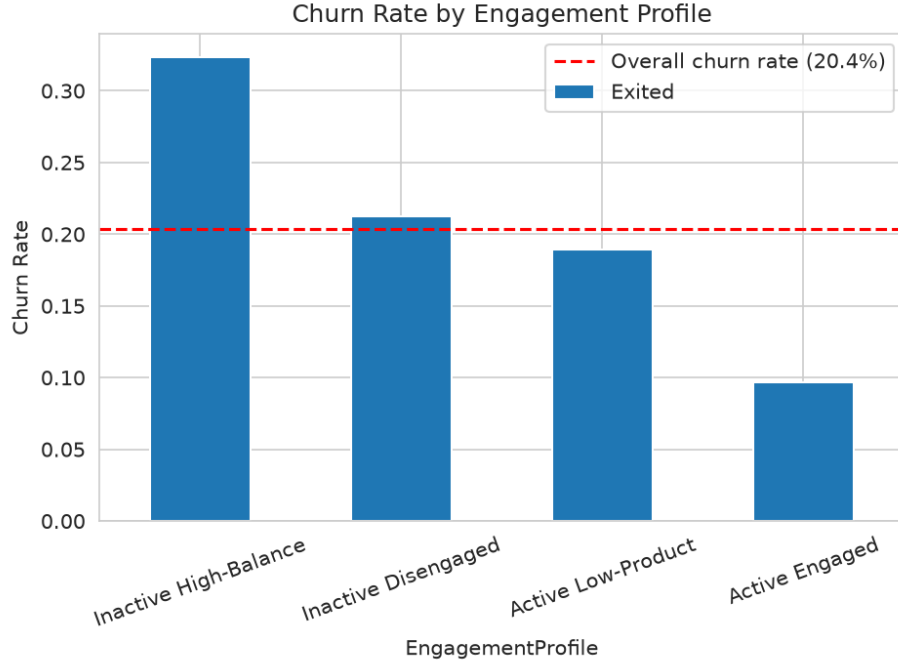


Figure 3: Churn rate by engagement profile, against the overall baseline.

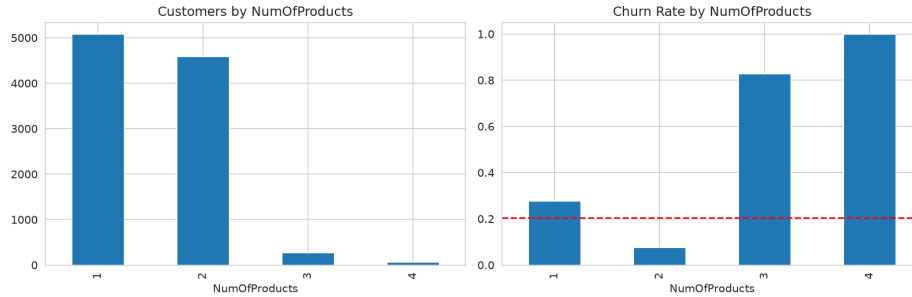


Figure 4: Churn rate by exact product count, showing the 3–4 product spike.

quartile, with active churn ranging from 11.0–18.8% versus 18.4–34.2% for inactive members across the same quartiles (Figure 5). Among inactive customers specifically, churn rose from 18.4% in the lowest balance quartile to 34.2% in the third quartile, before a slight dip in the highest quartile – reinforcing that wealth does not protect disengaged customers.

A live “at-risk premium” watchlist was constructed: customers who are inactive, in the top quartile for both balance and salary, and have not yet churned. This group comprises 220 customers (2.2% of the base), averaging €148,165 in balance and €176,746 in salary, and is nearly evenly split between France (42.3%) and Germany (41.4%) – a more geographically general risk group than the broader Inactive High-Balance segment.

A separate check for high-salary, zero-balance customers (a possible signal of a customer banking primarily elsewhere) found 891 such customers (8.9% of the base) with a churn rate of 14.8%, *below* the 20.37% baseline. This pattern appears stable rather than risky, likely reflecting customers who

use the bank for salary or transactions while holding savings elsewhere.

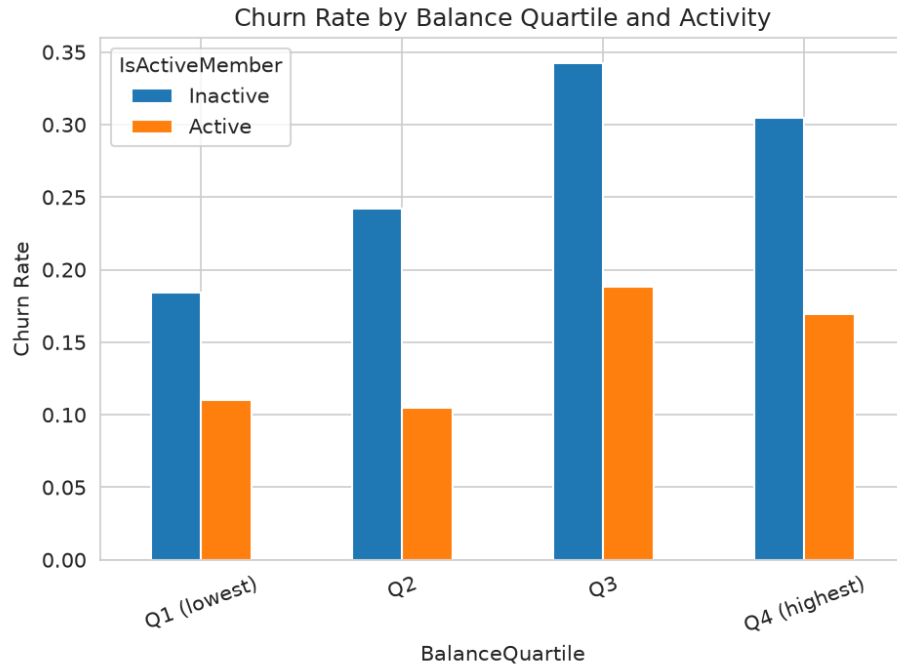


Figure 5: Churn rate by balance quartile, split by activity status.

4.5 Key Performance Indicators

Five KPIs were defined to summarize the behavioral drivers of retention (Table 3).

Table 3: Retention KPI definitions and results

KPI		Value	Definition
Engagement Retention Ratio		1.88	Inactive churn rate $\div$ active churn rate
Product Depth Index		0.80	Customer-weighted average retention score across product tiers
High-Balance Disengagement Rate		49.1%	Share of above-median-balance customers who are inactive
Credit Card Stickiness Score		0.006	Churn (no card) $-$ churn (has card)
Relationship Strength Index		$\bar{x} = 0.591$	Weighted composite: $0.4(\text{active}) + 0.3(\text{product fit}) + 0.1(\text{card}) + 0.2(\text{tenure, normalized})$

The Engagement Retention Ratio of 1.88 indicates inactive customers churn at nearly twice the rate of active customers. The Product Depth Index (0.80) masks the non-monotonic product pattern described in Section 4.3 and should always be read alongside the tier breakdown. The High-Balance Disengagement Rate shows that nearly half of above-median-balance customers are inactive, highlighting a segment that warrants proactive retention monitoring. The near-zero Credit Card Stickiness Score confirms card ownership is not a retention lever.

The Relationship Strength Index was sanity-checked on two criteria: it correlates negatively with churn ($r = -0.27$), and it ranks Germany as the geography with the lowest average score (0.576, versus 0.594 in France and 0.599 in Spain), consistent with Germany’s elevated churn rate throughout the analysis. These checks support the index’s internal consistency but do not constitute formal statistical validation.

5 Discussion

Taken together, these findings support a central thesis: **behavioral engagement provides a more actionable churn signal than financial wealth**. This holds at the single-variable level (Section 4.1), within the combined engagement classification (Section 4.2), and across every balance quartile (Section 4.4). Wealth alone offers no protection against churn once disengagement sets in – and in the Inactive High-Balance segment, may even coincide with elevated risk.

This is a claim about actionability, not raw predictive strength: age shows a stronger single-variable correlation with churn ($r = +0.29$) than activity status ($r = -0.16$, Section 4.1). Age and geography both interact with the engagement findings as partial confounders. None of this undermines the practical conclusion – a bank cannot change a customer’s age, but it can act on disengagement – though it does mean engagement-based retention strategy should be geography- and age-aware rather than applied uniformly.

The product utilization findings complicate the common cross-sell assumption that more products equal more loyalty. This holds only up to two products; beyond that, the relationship inverts sharply and for reasons the available data cannot fully explain. This should be treated as a priority area for further investigation, potentially requiring qualitative customer research the bank does not currently collect.

6 Recommendations

1. **Prioritize the Inactive High-Balance segment for re-engagement.** At 2,456 customers and a 32.3% churn rate, this is the largest high-risk engagement segment identified (the 3–4 product tier churns at a higher rate still, but is far smaller and addressed separately in Recommendation 3). Outreach should focus on reactivating usage (e.g., app engagement, transaction incentives) rather than further financial upselling.
2. **Deploy the 220-customer at-risk premium watchlist as a live intervention list.** These are the bank’s highest-value customers currently showing disengagement but not yet churned – the clearest opportunity for proactive retention outreach.
3. **Investigate the 3–4 product churn spike before continuing aggressive cross-sell targets.** Until the cause is understood, pushing customers toward a third or fourth product may be counterproductive to retention despite short-term revenue gains.
4. **Apply Germany-specific retention review.** Germany’s elevated churn appears across nearly every cut of the data (geography, engagement profile, product count). A market-specific investigation – pricing, competition, service quality – is warranted beyond what this behavioral dataset can explain.
5. **De-prioritize credit card promotion as a retention lever.** Card ownership shows no measurable effect on churn and should not be assumed to increase “stickiness.”

7 Limitations

- The dataset is a single-year snapshot; no time-series behavior (e.g., declining engagement trends prior to churn) could be analyzed.
- The Inactive High-Balance finding is partially confounded by geography and should not be read as a purely behavioral effect.
- The 3–4 product churn spike remains unexplained by the variables available in this dataset.
- Engagement profile thresholds (e.g., median balance split) were chosen analytically and are one reasonable choice among several; results may shift modestly under different threshold definitions.

8 Conclusion

This analysis highlights behavioral engagement as an important and actionable churn signal alongside demographic and financial factors. Engagement – not balance, salary, or product count in isolation – offers the bank the clearest lever for intervention, even though customer age shows the strongest single-variable correlation with churn overall. The recommendations above translate this into concrete, prioritized actions: a large re-engagement target (Inactive High-Balance), a small high-value watchlist (at-risk premium), an open investigation (the product spike), and a market-specific review (Germany). Together, these give the bank a behaviorally-grounded foundation for its retention strategy, moving beyond the generic, wealth-based targeting the original problem statement identified as insufficient.